



RKDF UNIVERSITY RANCHI

B.Sc. Botany

B.SC. BOTANY

NEP 2020

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B.Sc. Botany

SEMESTER I

Subject Code	Subject title	Credit
MJ01BOT	ALGAE AND MICROBIOLOGY	3

Unit 1: Introduction to microbial world

Microbial nutrition, growth and metabolism. Economic importance of viruses with reference to vaccine production, role in research, medicine and diagnostics, as causal organisms of plant diseases. Economic importance of bacteria with reference to their role in agriculture and industry (fermentation and medicine).

Unit 2: Viruses

Discovery, physiochemical and biological characteristics; classification (Baltimore), general structure with special reference to viroids and prions; replication (general account), DNA virus (T-phage), lytic and lysogenic cycle; RNA virus (TMV).

Unit 3: Bacteria

Discovery, general characteristics; Types- archaebacteria, eubacteria, wall-less forms (mycoplasma and spheroplasts); Cell structure; Nutritional types;

Reproduction vegetative, asexual and recombination (conjugation, transformation and transduction).

Unit 4: Algae

General characteristics; Ecology and distribution; range of thallus organization; Cell structure and components; cell wall, pigment system, reserve food (of only groups represented in the syllabus), flagella; methods of reproduction; Classification; criteria, system of Fritsch, and evolutionary classification of Lee (only upto groups); Significant contributions of important phycologists (F.E. Fritsch, G.M. Smith, R.N. Singh, T.V.



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Desikachary, H.D. Kumar, M.O.P. Iyengar). Role of algae in the environment, agriculture, biotechnology and industry.

Unit 5: Cyanophyta and Xanthophyta

Ecology and occurrence; Range of thallus organization; Cell structure; Reproduction, Morphology and life-cycle of Nostoc and Vaucheria.

Unit 6: Chlorophyta and Charophyta

General characteristics; Occurrence; Range of thallus organization; Cell structure; Reproduction. Morphology and life-cycles of Chlamydomonas, Volvox, Oedogonium, Coleochaete, Chara. Evolutionary significance of Prochloron.

Unit 7: Phaeophyta and Rhodophyta

Characteristics; Occurrence; Range of thallus organization; Cell structure; Reproduction. Morphology and life-cycles of Ectocarpus, Fucus and Polysiphonia.

Reference Books:

- Lee, R.E. (2008). Phycology, Cambridge University Press, Cambridge. 4th edition.
- Wiley JM, Sherwood LM and Woolverton CJ. (2013) Prescott's Microbiology. 9th Edition. McGraw Hill International.
- Kumar, H.D. (1999). Introductory Phycology. Affiliated East-West Press, Delhi.
- Pelczar, M.J. (2001) Microbiology, 5th edition, Tata McGraw-Hill Co, New Delhi.



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Subject Code	Subject title	Credit
MJL01(BOT)	ALGAE AND MICROBIOLOGY-LAB	1

PRACTICALS:

- Study of vegetative and reproductive structures of *Nostoc*, *Chlamydomonas* (electron micrographs), *Volvox*, *Oedogonium*, *Coleochaete*, *Chara*, *Vaucheria*, *Ectocarpus*, *Fucus* and *Polysiphonia*, *Prochloron* through electron micrographs, temporary preparations and permanent slides
- Diversity & Economic Importance of Microbes
- Electron micrographs/Models of viruses – T-Phage and TMV, Line drawings/ Photographs of Lytic and Lysogenic Cycle.
- Types of Bacteria to be observed from temporary/permanent slides/photographs. Electron micrographs of bacteria, binary fission, endospore, conjugation, root Nodule.
- Gram staining.
- Endospore staining with malachite green using the (endospores taken from soil bacteria)



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SEMESTER II

Subject Code	Subject title	Credit
MJ02BOT	BIOMOLECULES AND CELL BIOLOGY	3

Unit 1: Biomolecules

Types and significance of chemical bonds; Structure and properties of water; pH and buffers. Carbohydrates: Nomenclature and classification; Monosaccharides; Disaccharides; Oligosaccharides and polysaccharides. Lipids: Definition and major classes of storage and structural lipids; Fatty acids structure and functions; Essential fatty acids; Triacyl glycerols structure, functions and properties; Phosphoglycerides.

Proteins: Structure of amino acids; Levels of protein structure-primary, secondary, tertiary and quaternary; Protein denaturation and biological roles of proteins. Nucleic acids: Structure of nitrogenous bases; Structure and function of nucleotides; Types of nucleic acids; Structure of A, B, Z types of DNA; Types of RNA; Structure of tRNA. Laws of thermodynamics, concept of free energy, endergonic and exergonic reactions, coupled reactions, redox reactions. ATP: structure, its role as a energy currency molecule.

Unit 2: Enzymes

Structure of enzyme: holoenzyme, apoenzyme, cofactors, coenzymes and prosthetic group; Classification of enzymes; Features of active site, substrate specificity, mechanism of action (activation energy, lock and key hypothesis, induced - fit theory), Michaelis – Menten equation, enzyme inhibition and factors affecting enzyme activity.



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Unit 3: The cell

Cell as a unit of structure and function; Characteristics of prokaryotic and eukaryotic cells; Origin of eukaryotic cell (Endosymbiotic theory).

Unit 4: Cell wall and plasma membrane

Chemistry, structure and function of Plant cell wall. Overview of membrane function; fluid mosaic model; Chemical composition of membranes; Membrane transport – Passive, active and facilitated transport, endocytosis and exocytosis.

Unit 5: Cell organelles

Nucleus: Structure-nuclear envelope, nuclear pore complex, nuclear lamina, molecular organization of chromatin; nucleolus. Cytoskeleton: Role and structure of microtubules, microfilaments and intermediary filament. Chloroplast, mitochondria and peroxisomes: Structural organization; Function; Semiautonomous nature of mitochondria and chloroplast. Endomembrane system: Endoplasmic Reticulum – Structure, targeting and insertion of proteins in the ER, protein folding, processing; Smooth ER and lipid synthesis, export of proteins and lipids; Golgi Apparatus – organization, protein glycosylation, protein sorting and export from Golgi Apparatus; Lysosomes

Unit 6: Cell division

Phases of eukaryotic cell cycle, mitosis and meiosis; Regulation of cell cycle checkpoints, role of protein kinases.



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Reference Books:

- Nelson DL and Cox MM (2008) Lehninger Principles of Biochemistry, 5th Edition., W.H. Freeman and Company.
- Karp, G. (2010). Cell Biology, John Wiley & Sons, U.S.A. 6th edition.
- Hardin, J., Becker, G., Skliensmith, L.J. (2012). Becker's World of the Cell, Pearson Education Inc. U.S.A. 8th edition.



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Subject Code	Subject title	Credit
MJL02BOT	BIOMOLECULES AND CELL BIOLOGY-LAB	1

PRACTICALS:

- Study of cell and its organelles with the help of electron micrographs.
- Cytochemical staining of: DNA- Feulgen and cell wall in the epidermal peel of onion using Periodic Schiff's (PAS) staining technique.
- Study the phenomenon of plasmolysis and deplasmolysis.
- Study the effect of organic solvent and temperature on membrane permeability.
- Study different stages of mitosis and meiosis.
- Meiosis through temporary squash preparation.



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Subject Code	Subject title	Credit
MJ03BOT	Biology And Diversity of Fungi And Plant Pathology	3

Unit-1

Recent trends on the classification of fungi with reference to morphological and para morphological criteria; Comparative study of following sub-division; Mastigomycotina: Albugo, Peronospora, Plasmopora.

Unit-2

Comparative study of following sub-division; Zygomycotina: Mucor, Rhizopus, Syncephalastrum, Ascomycotina: Tapharina, Emericella, Penicillium, Chaetomium, Morchella.

Unit-3

Comparative study of following sub-division; Basidiomycotina: Puccinia, Melampsora, Ustilago, Polyporus, Cyathus, Deuteromycotina: Fusarium, Cercospora, Colletotrichum. Mushroom cultivation: Mycorrhizal application in agriculture and forestry; Fungal cytology and genetics: Heterothallism, heterokaryosis, parasexual cycle, mutation.

Unit-4

Symptomatology in fungal, bacterial and viral infection of plants, Etiology and control of the

Following crop diseases

1. Paddy: paddy blast, paddy blight
2. Wheat: Black stem rust, Bunt of wheat
3. Bajra: green ear and Ergot
4. Sugarcane: Red rot disease of sugarcane.
5. Ground nut: Tikka disease
6. Maize Smut



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Unit-5

Role of enzyme and toxins in pathogenesis; Disease control by physical, chemical and biological Methods, resistant varieties; Crop rotation, plant quarantines, seed certification.

Reference Books:

1. Agrios GN. 2005. Plant Pathology. 5th Ed. Academic Press, New York.
2. Heitefuss R & Williams PH. 1976. Physiological Plant Pathology. Springer Verlag, Berlin, New York.
3. Mehrotra RS & Aggarwal A. 2003. Plant Pathology. 2nd Ed. Oxford & IBH, New Delhi.
4. Singh RS. 2002. Introduction to Principles of Plant Pathology. Oxford & IBH, New Delhi.
5. Singh DP & Singh A. 2007. Disease and Insect Resistance in Plants. Oxford & IBH, New Delhi.
6. Upadhyay RK & Mukherjee KG. 1997. Toxins in Plant Disease Development and Evolving Biotechnology. Oxford & IBH, New Delhi.



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Subject Code	Subject title	Credit
MJL03BOT	BIOLOGY AND DIVERSITY OF FUNGI AND PLANT PATHOLOGY-LAB	1

1. Collection and study of algae mentioned in theory, identification up to generic level using
1. Algal monographs.
2. Preparation of synthetic medium and cultivation of algae, unialgal and axenic culture and their maintenance.
3. Collection, preservation of algal herbarium (10 specimens).
4. Preparation of pigments.
5. Staining techniques of cytology studies.
6. Study of diseased specimens of plants with reference to symptomatology.
7. Isolation, purification and single spore culture of pathogens.
8. Measurement of the activity of enzymes of fungal pathogens: Cellulase, Pectinases.
9. Laboratory testing of fungicides (systemic and non-systemic) against pathogenic fungi.
Preparation of culture media.
10. Preparation of culture media.



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SEMESTER III

Subject Code	Subject title	Credit
B.Sc. Botany	MYCOLOGY AND PHYTOPATHOLOGY	MJ04(BOT)

Unit 1: Introduction to true fungi

General characteristics; Affinities with plants and animals; Thallus organization; Cell wall composition; Nutrition; Classification.

Unit 2: Chytridiomycota and Zygomycota

Characteristic features; Ecology and significance; Thallus organisation; Reproduction; Life cycle with reference to Synchytrium, Rhizopus.

Unit 3: Ascomycota

General characteristics (asexual and sexual fruiting bodies); Ecology; Life cycle, Heterokaryosis and parasexuality; Life cycle and classification with reference to Saccharomyces, Aspergillus, Penicillium, Alternaria, Neurospora and Peziza.

Unit 4: Basidiomycota

General characteristics; Ecology; Life cycle and Classification with reference to black stem rust on wheat Puccinia (Physiological Specialization), loose and covered smut (symptoms only), Agaricus; Bioluminescence, Fairy Rings and Mushroom Cultivation.

Unit 5: Allied Fungi

General characteristics; Status of Slime molds, Classification; Occurrence; Types of plasmodia; Types of fruiting bodies.

Unit

6:

Oomycota

General characteristics; Ecology; Life cycle and classification with reference to Phytophthora, Albugo.



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Unit 7: Symbiotic associations

Lichen – Occurrence; General characteristics; Growth forms and range of thallus organization; Nature of associations of algal and fungal partners; Reproduction; Mycorrhiza-Ectomycorrhiza, Endomycorrhiza and their significance.

Reference Books:

1. Agrios, G.N. (1 997) Plant Pathology, 4th edition, Academic Press, U.K.
2. Alexopoulos, C.J., Mims, C.W., Blackwell, M. (1 996). Introductory Mycology, John Wiley & Sons (Asia) Singapore. 4th edition.
3. Webster, J. and Weber, R. (2007). Introduction to Fungi, Cambridge University Press, Cambridge. 3rd edition.



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Subject Code	Subject title	Credit
MJL04(BOT)	MYCOLOGY AND PHYTOPATHOLOGY-LAB	1

PRACTICALS:

1. Introduction to the world of fungi (Unicellular, coenocytic/septate mycelium, ascocarps & basidiocarps).
2. *Rhizopus*: study of asexual stage from temporary mounts and sexual structures through permanent slides.
3. *Aspergillus* and *Penicillium*: study of asexual stage from temporary mounts. Study of Sexual stage from permanent slides/photographs.
4. *Peziza*: sectioning through ascocarp.
5. *Alternaria*: Specimens/photographs and temporary mounts.
6. *Puccinia*: Herbarium specimens of Black Stem Rust of Wheat and infected Barberry leaves; sections/ mounts of spores on wheat and permanent slides of both the hosts.
7. *Agaricus*: Specimens of button stage and full-g r o w n mushroom; sectioning of gills of *Agaricus*, fairy rings and bioluminescent mushrooms to be shown.
8. Study of phaneroplasmodium from actual specimens and/or photograph. Study of *Stemonitis* sporangia
9. Albugo: Study of symptoms of plants infected with Albugo; asexual phase study through section/ temporary mounts and sexual structures through permanent slides.
10. Phytopathology: Herbarium specimens of bacterial diseases; Citrus Canker; Angular leaf spot of cotton, Viral diseases: TMV, Vein clearing, Fungal diseases: Early blight of potato, Black stem rust of wheat and White rust of crucifers.



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Subject Code	Subject title	Credit
MJ05BOT	ARCEGONIATE	3

Unit 1: Introduction

Unifying features of archegoniates; Transition to land habit; Alternation of generations.

Unit 2: Bryophytes

General characteristics; Adaptations to land habit; Classification; Range of thallus organization.

Unit 3: Type Studies- Bryophytes

Classification (up to family), morphology, anatomy and reproduction of Riccia, Marchantia, Pellia, Porella, Anthoceros, Sphagnum and Funaria; Reproduction and evolutionary trends in Riccia, Marchantia, Anthoceros and Funaria (developmental stages not included). Ecological and economic importance of bryophytes with special reference to Sphagnum.

Unit 4: Pteridophytes

General characteristics; Classification; Early land plants (Cooksonia and Rhynia).

Unit 5: Type Studies- Pteridophytes

Classification (up to family), morphology, anatomy and reproduction of Psilotum, Selaginella, Equisetum and Pteris (Developmental details not to be included). Apogamy, and apospory, heterospory and seed habit, telome theory, stelar evolution; Ecological and economic importance.



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Unit 6: Gymnosperm

General characteristics, classification (up to family), morphology, anatomy and reproduction of Cycas, Pinus and Gnetum (Developmental details not to be included); Ecological and economic importance

Reference Books

1. Vashistha, P.C., Sinha, A.K., Kumar, A. (2010). Pteridophyta. S. Chand. Delhi, India.
2. Bhatnagar, S.P. & Moitra, A. (1996). Gymnosperms. New Age International (P) Ltd Publishers, New Delhi, India.
3. Parihar, N.S. (1991). An introduction to Embryophyta: Vol. I. Bryophyta. Central Book Depot. Allahabad.
4. Raven, P.H., Johnson, G.B., Losos, J.B., Singer, S.R. (2005). Biology. Tata McGraw Hill, Delhi.
4. Vanderpoorten, A. & Goffinet, B. (2009) Introduction to Bryophytes. Cambridge University Press



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Subject Code	Subject title	Credit
MJL05BOT	ARCHEGONIATE-LAB	1

PRACTICALS:

Archegoniate **Riccia** – Morphology of thallus.

Marchantia- Morphology of thallus, whole mount of rhizoids & Scales, vertical section of thallus through Gemma cup, whole mount of Gemmae (all temporary slides), vertical section of Antheridiophore, Archegoniophore, longitudinal section of Sporophyte (all permanent slides).

Anthoceros- Morphology of thallus, dissection of sporophyte (to show stomata, spores, pseudoelaters, columella) (temporary slide), vertical section of thallus (permanent slide).

Pellia, **Porella**- Permanent slides.

Sphagnum- Morphology of plant, whole mount of leaf (permanent slide only).

Funaria- Morphology, whole mount of leaf, rhizoids, operculum, peristome, annulus, spores (temporary slides); permanent slides showing antheridial and archegonial heads, longitudinal section of capsule and protonema.

Psilotum- Study of specimen, transverse section of synangium (permanent slide).

Selaginella- Morphology, whole mount of leaf with ligule, transverse section of stem, whole mount of strobilus, whole mount of microsporophyll and megasporophyll (temporary slides), longitudinal section of strobilus (permanent slide).

Equisetum- Morphology, transverse section of internode, longitudinal section of strobilus, transverse section of strobilus, whole mount of sporangiophore, whole mount of spores (wet and dry) (temporary slide), transverse section of rhizome (permanent slide).

Pteris- Morphology, transverse section of rachis, vertical section of sporophyll, whole mount of sporangium, whole mount of spores (temporary slides), transverse section of rhizome, whole mount of prothallus with sex organs and young sporophyte (permanent slide).

Cycas- Morphology (coralloid roots, bulbil, leaf), whole mount of microsporophyll, transverse section of coralloid root, transverse section of rachis, vertical section of leaflet,



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vertical section of microsporophyll, whole mount of spores (temporary slides), longitudinal section of ovule, transverse section of root (permanent slide).

Pinus- Morphology (long and dwarf shoots, whole mount of dwarf shoot, male and female cones), transverse section of Needle, transverse section of stem, longitudinal section of / transverse section of male cone, whole mount of microsporophyll, whole mount of Microspores (temporary slides), longitudinal section of female cone, tangential longitudinal section & radial longitudinal sections stem (permanent slide).

Gnetum- Morphology (stem, male & female cones), transverse section of stem, vertical section of ovule (permanent slide)



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SEMESTER IV

Subject Code	Subject title	Credit
MJ06BOT	PLANT TAXONOMY & ECONOMIC BOTANY	3

Plant Taxonomy

Unit 1: Introduction to Plant Taxonomy 1. Fundamental components of taxonomy (identification, nomenclature, classification) 2. Taxonomic resources: Herbarium- functions & important herbaria, Botanical gardens, Flora. 3. Botanical Nomenclature- Principles and rules of ICN (ranks and names; principle of priority, binomial system; type method (Typification), author citation and valid-publication).

Unit 2: Taxonomic hierarchy, Types of classification and Evidences 1. Concept of taxa (family, genus, species); Categories and taxonomic hierarchy; Species concept (taxonomic, biological, evolutionary). 2. Types of classification- Artificial, Natural and Phylogenetic. 3. Bentham & Hooker's system of classification (up to series) - merits and demerits. 4. Engler & Prantle's system of classification (up to series) - merits and demerits. 5. Hutchinson classification with its - merits and demerits. 6. Angiosperm phylogeny group (APG IV) system classification. 7. Taxonomic evidences from palynology, cytology, phytochemistry & Molecular biology data (Protein and Nucleic acid homology).

Unit 3: Plant Systematics 1. Diagnostic characteristics, Systematic Phylogeny and economic importance of families: Ranunculaceae, Brassicaceae, Apocynaceae, Apiaceae, Asteraceae, Solanaceae, Lamiaceae, Euphorbiaceae. Liliaceae, Poaceae.

Unit 4: Modern trends in Plant taxonomy: 1. Phenetics and Cladistics: Brief idea on Phenetics, Numerical taxonomy- methods, Operational Taxonomic Units (OUT's), 2. Cladistics- construction of dendrogram and primary analysis; Monophyletic, polyphyletic and paraphyletic groups; Plesiomorphy and apomorphy. 3. Origin and evolution of angiosperms; Methods of illustrating evolutionary relationship (phylogenetic tree, cladogram).



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Economic Botany

Unit 1: Introduction, Research centres, Concept of centres of origin, their importance with reference to Vavilov's work. Study of following economically important plants with special reference to Jharkhand:

Unit 2: Cereals and Millets: Wheat, Rice, Ragi and Jowar -Origin, morphology, uses

Unit 3: Pulses & Vegetables General account with special reference to Gram, soybean and Potato

Unit 4: Spices: General account with special reference to clove, black pepper, cinnamon, Ginger and Turmeric (Botanical name, family, part used, morphology and uses)

Unit 5: Beverages Tea and Coffee (morphology, processing, uses)

Unit 6: Oils and Sugar General Description with special reference to groundnut and sugarcane

Unit 7: Timber and Fiber and Yielding Plants General description (Botanical name, family, parts used, morphology and uses)

Unit 8: Medicinal Plants Brief account of Ocimum, Tinospora, Aloe, Rauvolfia, Emblica and Catharanthus (Botanical name, family, part used, morphology and uses)

Reference Books:

1. Singh, (2012). Plant Systematics: Theory and Practice Oxford & IBH Pvt. Ltd., New Delhi. 3rd edition.
2. Jeffrey, C. (1982). An Introduction to Plant Taxonomy. Cambridge University Press, Cambridge.
3. Judd, W.S., Campbell, C.S., Kellogg, E.A., Stevens, P.F. (2002). Plant Systematics-A Phylogenetic Approach. Sinauer Associates Inc., U.S.A. 2nd edition.
4. Maheshwari, J.K. (1963). Flora of Delhi. CSIR, New Delhi.
5. Radford, A.E. (1986). Fundamentals of Plant Systematics. Harper and Row, New York.
6. Kochhar, S.L. (2012). Economic Botany in Tropics, MacMillan & Co. New Delhi, India.



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7. Wickens, G.E. (2001). Economic Botany: Principles & Practices. Kluwer Academic Publishers, The Netherlands.



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Subject Code	Subject title	Credit
MJL06BOT	PLANT TAXONOMY & ECONOMIC BOTANY-Lab	1

PRACTICALS:

Plant Taxonomy & Economic Botany

1. Systematic study of locally available plants belonging to the families prescribed in theory syllabus with reference to vegetative and floral characters of the following families (Description, V.S. flower, section of ovary, floral diagram/s, floral formula/e and systematic position according to Bentham & Hooker's system of classification)
2. Field visit (local)
3. Mounting of a properly dried and pressed specimen of any wild plant with herbarium label (to be submitted in the record book).
4. Study of economically important plants: Wheat, Rice, Gram, Soybean, Potato, Black pepper, Clove, Cinnamon, Ginger, Turmeric, Tea, Coffee, Cotton, Groundnut, Sugarcane and Medicinal plants through specimens, sections.



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Subject Code	Subject title	Credit
MJ07BOT	DIVERSITY OF PLANTS	3

DIVERSITY OF PLANTS

Unit 1: Microbes Viruses – Discovery, general structure, replication (general account), DNA virus (T-phage); Lytic and lysogenic cycle, RNA virus (TMV); Economic importance; Bacteria – Discovery, General characteristics and cell structure; Reproduction – vegetative, asexual and recombination (conjugation, transformation and transduction); Economic importance.

Unit 2: Algae General characteristics; Ecology and distribution; Range of thallus organization and reproduction; Classification of algae; Morphology and life-cycles of the following: Nostoc, Oedogonium, Vaucheria, Ectocarpus, Polysiphonia. Economic importance of algae

Unit 3: Fungi Introduction- General characteristics, ecology and significance, range of thallus organization, cell wall composition, nutrition, reproduction and classification; True Fungi- General characteristics, ecology and significance, life cycle of Penicillium, Puccinia, Ustilago, Alternaria; Symbiotic Associations Lichens: General account of Mycorrhiza: ectomycorrhiza and endomycorrhiza and their significance.

Unit 4: Introduction to Archegoniate Identifying features of archegoniates, Transition to land habit, Alternation of generations.

Unit 5: Bryophytes General characteristics, adaptations to land habit, Classification, Range of thallus organization. Classification (up to family), morphology, anatomy and reproduction of Marchantia and Funaria. (Developmental details not to be included). Ecology and economic importance of bryophytes with special reference to Sphagnum.



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Unit 6: Pteridophytes General characteristics, classification, Early land plants Fossil and Fossilization process (Rhynia). Classification (up to family), morphology, anatomy and reproduction of Lycopodium, Equisetum and Pteris. Heterospory and seed habit, stelar evolution.

Unit 7: Gymnosperms General characteristics, classification (up to family), morphology, anatomy and reproduction of Cycas and Pinus. Ecological and economical importance.

Reference Books:

1. Kumar, H.D. (1999). Introductory Phycology. Affiliated East-West. Press Pvt. Ltd. Delhi. 2nd edition.
2. Tortora, G.J., Funke, B.R., Case, C.L. (2010). Microbiology: An Introduction, Pearson Benjamin Cummings, U.S.A. 10th edition.
2. Sethi, I.K. and Walia, S.K. (2011). Text book of Fungi & Their Allies, MacMillan Publishers Pvt. Ltd., Delhi.
3. Alexopoulos, C.J., Mims, C.W., Blackwell, M. (1996). Introductory Mycology, John Wiley and Sons (Asia), Singapore. 4th edition.
4. Raven, P.H., Johnson, G.B., Losos, J.B., Singer, S.R., (2005). Biology. Tata McGraw Hill, Delhi, India.
5. Vashishta, P.C., Sinha, A.K., Kumar, A., (2010). Pteridophyta, S. Chand. Delhi, India.
6. Bhatnagar, S.P. and Moitra, A. (1996). Gymnosperms. New Age International (P) Ltd Publishers, New Delhi, India.
7. Parihar, N.S. (1991). An introduction to Embryophyta. Vol. I. Bryophyta. Central Book Depot, Allahabad.
8. Carson, R. 2002. Silent Spring. Houghton Mifflin Harcourt.



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Subject Code	Subject title	Credit
MJL07BOT	DIVERSITY OF PLANTS-LAB	1

PRACTICALS:

1. EMs/Models of viruses – T-Phage and TMV, Line drawing/Photograph of Lytic and Lysogenic Cycle. 2. Types of Bacteria from temporary/permanent slides/photographs; EM bacterium; Binary Fission; Conjugation; Structure of root nodule.
2. Gram staining.
3. Morphology and structural details of forms belonging to Algae, Fungi, Bryophytes, Pteridophytes and Gymnosperm prescribed in the syllabus and their temporary/permanent stained microscopic slide preparation and studies.
4. Comments upon the spots
5. Vive-voce
6. Field study report
7. Class records, Herbarium, Charts, Model etc. 9. Study of instruments used to measure microclimatic variables: Soil thermometer, maximum and minimum thermometer, anemometer, hygrometer, rain gauge and lux meter.



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Subject Code	Subject title	Credit
MJ08BOT	CROP PATHOLOGY	3

Unit-1

Introduction to crop pathology: Classification of plant diseases based on;

- (a) Major causal agents - biotic and abiotic,
- (b) General symptoms.

Unit-2 Process of infection and pathogenesis:

- (a) Penetration and entry of pathogen into host tissue – mechanical, physiological and enzymatic.
- (b) Host-parasite interaction, enzymes and toxins in pathogenesis.

Unit-3

Defense mechanism in plants: Pre-existing structural and biochemical defense mechanisms, lack of essential nutrients. Induced structural and biochemical defense mechanisms, inactivation

of pathogen enzymes and toxins, altered biosynthetic pathways.

Unit-4

Transmission of plant disease: Spread and transmission of plant diseases by wind, water, seeds

and vectors. Plant disease management: Exclusion, eradication and protection. Chemical means

of disease control – common fungicides, antibiotics and nematicides. Biological means of disease control. Biotechnological approaches to disease resistance: Fungi in agricultural biotechnology, control of fungal plant pathogens by mycofungicides; Transgenic approaches to disease resistance.

Unit-5- Major diseases in plants

- (a) Cereals: Rice - blast disease, bacterial blight; Wheat - black rust disease.
- (b) Vegetables: Chilly - leaf spot; Ladies finger - vein clearing disease.



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- (c) Fruits: Banana - bacterial leaf blight, leaf spot; Mango - Anthracnose; Citrus - bacterial canker; Papaya – mosaic.
- (d) Spices: Ginger - rhizome rot; Pepper - quick wilt; Cardamom - marble mosaic disease.
- (e) Oil seeds: Coconut - grey leaf spot, bud rot disease.

References

1. K S Bilgrami, H C Dube. A text book of modern plant pathology.
2. Gareth Johnes. Plant pathology: principles and practice.
3. R S Mehrotra. Plant Pathology. 4. M N Kamat. Practical plant pathology.



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MJL08BOT	Crop Pathology-Lab	1

1. Isolation of Bacillus and Rhizobium spp from soil and nodules.
2. Various methods of bacterial staining to study cell wall, endospore, capsule and flagella.
3. Identification of important genera by using biochemical tests: Escherichia. Azatobactor,
4. Staphylococcus, Bacillus, Pseudomonas, Rhizobium, Streptomyces, Xanthomonas.
5. Construction of bacterial growth curve.
6. Quantitative estimation of bacteria in milk
7. Make suitable micro preparations and identify the diseases mentioned with due emphasis on symptoms and causative organisms.
8. Isolation of pathogens from diseased tissues (leaf, stem and fruit) by serial dilution method.



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SEMESTER - V

Subject Code	Subject title	Credit
MJ09BOT	CELL BIOLOGY & GENETICS	3

Cell Biology

Unit1: The cell Cell as a unit of structure and function; Characteristics of prokaryotic and eukaryotic cells; Origin of eukaryotic cell (Endosymbiotic theory).

Unit 2: Cell wall and plasma membrane Chemistry, structure and function of Plant cell wall. Overview of membrane function; fluid mosaic model; Chemical composition of membranes; Membrane transport – Passive, active and facilitated transport, endocytosis and exocytosis.

Unit 3: Cell organelles Nucleus: Structure-nuclear envelope, nuclear pore complex, nuclear lamina, molecular organization of chromatin; nucleolus. Cytoskeleton: Role and structure of microtubules, microfilaments and intermediary filament. Chloroplast, mitochondria and peroxisomes: Structural organization; Function; Semiautonomous nature of mitochondria and chloroplast. Endomembrane system: Endoplasmic Reticulum – Structure, targeting and insertion of proteins in the ER, protein folding, processing; Smooth ER and lipid synthesis, export of proteins and lipids; Golgi Apparatus – organization, protein glycosylation, protein sorting and export from Golgi Apparatus; Lysosomes.

Unit 4: Cell division Phases of eukaryotic cell cycle, mitosis and meiosis; Regulation of cell cycle- checkpoints, role of protein kinases

Genetics

Unit 1: Mendelian genetics and its extension Mendelism: History; Principles of inheritance; Chromosome theory of inheritance; Autosomes and sex chromosomes; Probability and pedigree analysis; Incomplete dominance and codominance; Multiple alleles, Lethal alleles, Epistasis, Pleiotropy, Recessive and Dominant traits, Penetrance and Expressivity, Numericals; Polygenic inheritance.

Unit 2: Extrachromosomal Inheritance Chloroplast mutation: Variegation in Four o'clock plant; Mitochondrial mutations in yeast; Maternal effects-shell coiling in snail; Infective heredity-Kappa particles in Paramecium.



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Unit 3: Linkage, crossing over and chromosome mapping Linkage and crossing over- Cytological basis of crossing over; Recombination frequency, two factor and three factor crosses; Interference and coincidence; Numericals based on gene mapping; Sex Linkage. (7 lectures)

Unit 4: Variation in chromosome number and structure Deletion, Duplication, Inversion, Translocation, Euploidy and Aneuploidy

Unit 5: Gene mutations Types of mutations; Molecular basis of Mutations; Mutagens – physical and chemical (Base analogs, deaminating, alkylating and intercalating agents); Detection of mutations: ClB method. Role of Transposons in mutation. DNA repair mechanisms.

Unit 6: Fine structure of gene Classical vs molecular concepts of gene; Cis-Trans complementation test for functional allelism; Structure of Phage T4, rII Locus.

Unit 7. Population and Evolutionary Genetics Allele frequencies, Genotype frequencies, Hardy-Weinberg Law, role of natural selection, mutation, genetic drift. Genetic variation and Speciation.

Reference Books:

1. Karp, G. (2010). Cell Biology, John Wiley & Sons, U.S.A. 6th edition.
2. Hardin, J., Becker, G., Skliensmith, L.J. (2012). Becker's World of the Cell, Pearson Education Inc. U.S.A.
3. Cooper, G.M. and Hausman, R.E. (2009) The Cell: A Molecular Approach. 5th edition. ASM Press & Sunderland, Washington, D.C.; Sinauer Associates, MA.
4. Becker, W.M., Kleinsmith, L.J., Hardin. J. and Bertoni, G. P. (2009) The World of the Cell. 7th edition. Pearson Benjamin Cummings Publishing, San Francisco



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Subject Code	Subject title	Credit
MJL09BOT	CELL BIOLOGY & GENETICS-Lab	1

PRACTICALS:

1. Study of cell and its organelles with the help of electron micrographs.
2. Cytochemical staining of: DNA- Feulgen and cell wall in the epidermal peel of onion using Periodic Schiff's (PAS) staining technique.
3. Study the phenomenon of plasmolysis and deplasmolysis.
4. Study the effect of organic solvent and temperature on membrane permeability.
5. Study different stages of mitosis and meiosis.
6. Meiosis through temporary squash preparation.
7. Mendel's laws through seed ratios. Laboratory exercises in probability and chi- square.
8. Chromosome mapping using point test cross data.
9. Pedigree analysis for dominant and recessive autosomal and sex-linked traits.
10. Incomplete dominance and gene interaction through seed ratios6. (9:7, 9:6:1, 13:3, 15:1, 12:3:1, 9:3:4).
11. Blood Typing: ABO groups & Rh factor.
12. Study of aneuploidy: Down's, Klinefelter's and Turner's syndromes.
13. Photographs/Permanent Slides showing Translocation Ring, Laggards and Inversion Bridge.
14. Study of human genetic traits: Sickle cell anemia, Xeroderma Pigmentosum, Albinism, red-green Colour blindness, Widow's peak, rolling of tongue, Hitchhiker's thumb and Attached ear lobe.



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Subject Code	Subject title	Credit
MJ10BOT	PLANT PHYSIOLOGY & BIOCHEMISTRY	3

Unit 1: Plant-water relations Water Potential and its components, mechanism of water absorption-active and passive absorption, aquaporins, pathway of water movement, symplast, apoplast, transmembrane pathways, root pressure, guttation. Ascent of sap-cohesion-tension theory. Transpiration and factors affecting transpiration, antitranspirants, mechanism of stomatal movement.

Unit 2: Mineral nutrition Essential and beneficial elements, macro and micronutrients, methods of study and use of nutrient solutions, criteria for essentiality, mineral deficiency symptoms, roles of essential elements, chelating agents.

Unit 3: Nutrient Uptake Soil as a nutrient reservoir, transport of ions across cell membrane, passive absorption, electrochemical gradient, facilitated diffusion, active absorption, role of ATP, carrier systems, proton ATPase pump and ion flux, uniport, co-transport, symport, antiport.

Unit 4: Translocation in the phloem Experimental evidence in support of phloem as the site of sugar translocation. Pressure-Flow Model; Phloem loading and unloading; Source-sink relationship.

Unit 5: Photosynthesis Photosynthesis as a chemical process – Light and Dark reaction; mechanism of absorption of light. The pigment system – PS I and PS II. Phosphorylation – Electron Transport System and Photophosphorylation (Cyclic and Non-cyclic). Hatch and Slack Pathway. CAM Cycle; Significance of C₄ cycle and CAM. Factors affecting rate of photosynthesis. Significance of photosynthesis.

Unit 6: Respiration Types of respiration, mechanism (Glycolysis). Krebs's cycle: Electron Transport System, Oxidative phosphorylation, fermentation. Factors affecting rate of respiration. Photorespiration.

Unit 7: Plant growth regulators Discovery, chemical structure, biosynthesis, bioassays and physiological roles of Auxin, Gibberellins, Cytokinin, Abscissic acid, Ethylene, Brassinosteroids and Jasmonic acid.(6 lectures) **Unit 8:** Physiology of flowering



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Photoperiodism, flowering stimulus, florigen concept, vernalization, seed germination, seed dormancy.

Unit 9: Phytochrome, cryptochromes and phototropins Discovery, chemical nature and structure, role in photomorphogenesis, low energy responses (LER) and high irradiance responses (HIR), mode of action.

Unit 10: Biomolecules Types and significance of chemical bonds; Structure and properties of water; pH and buffers. Carbohydrates: Nomenclature and classification; Monosaccharides; Disaccharides; Oligosaccharides and polysaccharides and its significance. Lipids: Definition and major classes of storage and structural lipids; Fatty acids structure and functions; Essential fatty acids; Triacyl glycerols structure, functions and properties; Phosphoglycerides. Proteins: Structure of amino acids; Levels of protein structure-primary, secondary, tertiary and quaternary; Protein denaturation and biological roles of proteins. Nucleic acids: Structure of nitrogenous bases; Structure and function of nucleotides; Types of nucleic acids; Structure of A, B, Z types of DNA; Types of RNA; Structure of tRNA.

Unit 11: Bioenergetics Laws of thermodynamics, concept of free energy, endergonic and exergonic reactions, coupled reactions, redox reactions. ATP: structure, its role as an energy currency molecule. **Unit 12:** Enzymes Structure of enzyme: holoenzyme, apoenzyme, cofactors, coenzymes and prosthetic group; Classification of enzymes; Features of active site, substrate specificity, mechanism of action (activation energy, lock and key hypothesis, induced - fit theory), Michaelis – Menten equation, enzyme inhibition and factors affecting enzyme activity.

Unit 13: Vitamins General characteristics of vitamins, hormones. Nomenclature and classification of vitamins and its significance.

Reference Books:

1. Hopkins, W.G. and Huner, A. (2008). Introduction to Plant Physiology. John Wiley and Sons. U.S.A. 4th edition.
2. Taiz, L., Zeiger, E., Møller, I.M. and Murphy, A (2015). Plant Physiology and Development. Sinauer Associates Inc. USA. 6th edition.
3. Bajracharya D. (1999). Experiments in Plant Physiology-A Laboratory Manual. Narosa Publishing House, New Delhi.



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4. Campbell, MK (2012) Biochemistry, 7th ed., Published by Cengage Learning
5. Campbell, PN and Smith AD (2011) Biochemistry Illustrated, 4th ed., Published by Churchill Livingstone



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Subject Code	Subject title	Credit
MJL10BOT	PLANT PHYSIOLOGY & BIOCHEMISTRY- Lab	1

PRACTICALS:

Plant Physiology & Biochemistry

1. Determination of osmotic potential of plant cell sap by plasmolytic method.
2. Determination of water potential of given tissue (potato tuber) by weight method.
3. Study of the effect of wind velocity and light on the rate of transpiration in excised twig/leaf.
4. Calculation of stomatal index and stomatal frequency from the two surfaces of leaves of mesophyte and xerophyte.
5. To study the phenomenon of seed germination (effect of light).
6. To study the effect of different concentrations of IAA on Avena coleoptile elongation (IAA Bioassay).
7. To study the induction of amylase activity in germinating barley grains.
8. Perform rate of photosynthesis and oxygen evolution by Wilmot's bubbler.
9. Perform Moll's experiment.
10. Demonstration experiments 1. To demonstrate suction due to transpiration. 2. Fruit ripening/Rooting from cuttings (Demonstration). 3. Bolting experiment/Avena coleoptile bioassay (demonstration). 4. Qualitative tests for carbohydrates, reducing sugars, non-reducing sugars, lipids and proteins. 5. Study of plant cell structure with the help of epidermal peel mount of Onion/Rhoeo/Crinum. 6. Demonstration of the phenomenon of protoplasmic streaming in Hydrilla leaf. 7. Measurement of cell size by the technique of micrometry. 8. Counting the cells per unit volume with the help of haemocytometer. (Yeast/pollen grains).



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Subject Code	Subject title	Credit
MJ11BOT	TAXONOMY, ANATOMY & EMBRYOLOGY	3

Unit 1: Introduction to plant taxonomy Identification, Classification, Nomenclature.

Unit 2: Identification Functions of Herbarium, important herbaria and botanical gardens of the world and India. Elementary idea of documentation of Flora.

Unit 3: Family description Magnoliaceae, Ranunculaceae, Lamiaceae, Poaceae.

Unit 4: Botanical nomenclature Principles and rules (ICBN); ranks and names; binominal system, typification, author citation, valid publication, rejection of names, principle of priority and its limitations.

Unit 5: Classification Types of classification-artificial, natural and phylogenetic. Bentham and Hooker (upto series), Engler and Prantl (upto series).

Unit 6: Biometrics, numerical taxonomy and cladistics Phenograms, cladograms definitions and differences.

Unit 7: Meristematic and permanent tissues Root and shoot apical meristems; Simple and complex tissue.

Unit 8: Organs Structure of dicot and monocot root stem and leaf.

Unit 9: Secondary Growth Vascular cambium – structure and function, seasonal activity. Secondary growth in root and stem, Wood (heartwood and sapwood).

Unit 10: Adaptive and protective systems Epidermis, cuticle, stomata; General account of adaptations in xerophytes and hydrophytes.

Unit 11: Structural organization of flower Structure of anther and pollen; Structure and types of ovules; Types of embryo sacs, organization and ultrastructure of mature embryo sac.

Unit 12: Pollination and fertilization Pollination mechanisms and adaptations; Double fertilization; Seed-structure.

Unit 13: Embryo and endosperm Endosperm types, structure and functions; Dicot and monocot embryo.

Unit 14: Apomixis and polyembryony Definition, types and practical applications.



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Reference Books:

1. Simpson, M.G. (2006). Plant Systematics. Elsevier Academic Press, San Diego, CA, U.S.A.
2. Singh, G. (2012). Plant Systematics: Theory and Practice. Oxford & IBH Pvt. Ltd., New Delhi. 3rd edition.
3. Bhojwani, S.S. & Bhatnagar, S.P. (2011). Embryology of Angiosperms. Vikas Publication House Pvt. Ltd. New Delhi. 5th edition.
4. Mauseth, J.D. (1988). Plant Anatomy. The Benjamin/Cummings Publisher, USA.



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Subject Code	Subject title	Credit
MJL11BOT	TAXONOMY, ANATOMY & EMBRYOLOGY-Lab	1

PRACTICALS:

1. Quantitative analysis of herbaceous vegetation in the college campus for frequency and comparison with Raunkiaer's frequency distribution law
2. Study of vegetative and floral characters of the following families (Description, V.S. flower, section of ovary, floral diagram/s, floral formula/e and systematic position according to Bentham & Hooker's system of classification): Brassicaceae -Brassica; Asteraceae - Launaea, Ageratum, Eclipta/Tridax; Solanaceae -Solanum nigrum; Apocyanaceae: Catharanthus, Thevetia, Lantana, Verbenaceae Liliaceae - Lilium/Allium.
3. Mounting of a properly dried and pressed specimen of any wild plant with herbarium label (to be submitted in the record book).
4. Study of meristems through permanent slides and photographs.
5. Tissues (parenchyma, collenchyma and sclerenchyma); Macerated xylary elements, Phloem (Permanent slides, photographs)
6. Stem: Monocot: Zea mays; Dicot: Helianthus; Secondary: Helianthus (only Permanent slides).
7. Root: Monocot: Zea mays; Dicot: Helianthus; Secondary: Helianthus (only Permanent slides).
8. Leaf: Dicot and Monocot leaf (only Permanent slides).
9. Adaptive anatomy: Xerophyte (Nerium leaf); Hydrophyte (Hydrilla stem).
10. Structure of anther (young and mature), tapetum (amoeboid and secretory) (Permanent slides).
11. Types of ovules: anatropous, orthotropous, circinotropous, amphitropous/ campylotropous.
12. Female gametophyte: Polygonum (monosporic) type of Embryo sac Development (Permanent slides/photographs).
13. Ultrastructure of mature egg apparatus cells through electron micrographs.



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14. Pollination types and seed dispersal mechanisms (including appendages, aril, caruncle)
(Photographs and specimens).
15. Dissection of embryo/endosperm from developing seeds.



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Subject Code	Subject title	Credit
SIP	INTERNSHIP/PROJECT WORK	4

Overall Internship/ Apprenticeship/ Project may be evaluated under the following heads:

- ☐ Motivation for the choice of topic
- ☐ Project dissertation design
- ☐ Methodology and Content depth
- ☐ Results and Discussion
- ☐ Future Scope & References
- ☐ Presentation style
- ☐ Viva-voce

Internship/ Apprenticeship/ Project

Each student has to submit two copies of the Internship/ Apprenticeship/ Project work duly forwarded by the HOD of Department concerned. The forwarded copies will be submitted in the Department of Botany, RKDF University, for evaluation (Seven days before the seminar).

The paper will consist of

- (a) Field work/Lab work related to the project.
- (b) Preparation of dissertation based on the work undertaken.
- (c) Presentation of project work in the seminar on the assigned topic in the U.G. Department of Botany, RKDF University, Ranchi & open viva thereon.

- ☐ Student alone or in a group of not more than five, shall undertake one Project approved by the Subject Teacher/H.O.D. of the Department/College concerned. The progress of the Project shall be monitored by the faculty members at regular Intervals.

OR

- ☐ Paper presentation on 'Topic Provided' and group discuss.



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SEMESTER VI

Subject Code	Subject title	Credit
MJ12BOT	MOLECULAR BIOLOGY	3

Unit 1: Nucleic acids: Carriers of genetic information. Introduction, DNA as the carrier of genetic information (Griffith's, McLeod & McCarty experiment).

Unit 2. The Structures of DNA and RNA / Genetic Material DNA Structure: Watson and Crick model, Salient features of double helix, Types of DNA, Types of genetic material, denaturation and renaturation, Organization of DNA- Prokaryotes, Viruses, Eukaryotes. RNA Structure Organelle DNA -- mitochondria and chloroplast DNA. The Nucleosome Chromatin structure- Euchromatin, Heterochromatin- Constitutive and Facultative heterochromatin.

Unit 3: The replication of DNA Chemistry of DNA synthesis (Kornberg's discovery); General principles – bidirectional, semiconservative, and semi discontinuous replication, RNA priming; Various models of DNA replication, including rolling circle, θ (theta) mode of replication, replication of linear ds-DNA, replication of the 5' end of linear chromosome; Enzymes involved in DNA replication.

Unit 4: Central dogma and genetic code Key experiments establishing-The Central Dogma (Adaptor hypothesis and discovery of m-RNA template), Genetic code (deciphering and salient features)

Unit 5: Transcription Transcription in prokaryotes and eukaryotes. Principles of transcriptional regulation; Prokaryotes: Regulation of lactose metabolism and tryptophan synthesis in *E. coli*. Eukaryotes: transcription factors, heat shock proteins, steroids and peptide hormones; Gene silencing.

Unit 6: Processing and modification of RNA Split genes-concept of introns and exons, removal of introns, spliceosome machinery, splicing pathways, group I and group II intron splicing, alternative splicing eukaryotic mRNA processing (5' cap, 3' polyA tail); Ribozymes; RNA editing and mRNA transport.

Unit 7: Translation Ribosome structure and assembly, mRNA; Charging of tRNA, aminoacyl tRNA synthetases; Various steps in protein synthesis, proteins involved in initiation, elongation



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and termination of polypeptides; Fidelity of translation; Inhibitors of protein synthesis; post-translational modifications of proteins.

Reference Books:

1. Watson J.D., Baker, T.A., Bell, S.P., Gann, A., Levine, M., Losick, R. (2007). Molecular Biology of the Gene, Pearson Benjamin Cummings, CSHL Press, New York, U.S.A. 6th edition.
2. Snustad, D.P. and Simmons, M.J. (2010). Principles of Genetics. John Wiley and Sons Inc., U.S.A. 5th ed.
3. Klug, W.S., Cummings, M.R., Spencer, C.A. (2009). Concepts of Genetics. Benjamin Cummings. U.S.A. 9th edition.
4. Russell, P. J. (2010). i-Genetics- A Molecular Approach. Benjamin Cummings, U.S.A. 3rd edition.
5. Griffiths, A.J.F., Wessler, S.R., Carroll, S.B., Doebley, J. (2010). Introduction to Genetic Analysis. W. H. Freeman and Co., U.S.A. 10th edition.



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Subject Code	Subject title	Credit
MJL12BOT	MOLECULAR BIOLOGY -Lab	1

PRACTICALS:

1. Preparation of LB medium and raising *E.Coli*.
2. Isolation of genomic DNA from *E.Coli*.
3. DNA isolation from cauliflower head.
4. DNA estimation by diphenylamine reagent/UV Spectrophotometry.
5. Study of DNA replication mechanisms through photographs (Rolling circle, Theta replication and semi-discontinuous replication).
6. Study of structures of prokaryotic RNA polymerase and eukaryotic RNA polymerase II through photographs.
7. Photographs establishing nucleic acid as genetic material (Messelson and Stahl's, Avery et al, Griffith's, Hershey & Chase's and Fraenkel & Conrat's experiments)
8. Study of the following through photographs: Assembly of Spliceosome machinery; Splicing mechanism in group I & group II introns; Ribozyme and Alternative splicing



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Subject Code	Subject title	Credit
MJ13BOT	COMPUTATIONAL BIOLOGY & RESEARCH METHODOLOGY FOR PLANTS	3

Computational Biology

1. Computer assisted drug design- concept, methods and practical approaches.
2. Diagrammatic, graphical and tabular representations of data; measures of central tendency, dispersion, skewness and kurtosis.
3. Basic concepts of hypothesis testing, two kinds of error, level significance, p value, t- Test for mean and difference between two means, partial t-test., and Chi square test for goodness of fit.
4. Application of GIS

Research Methodology for Plants

INSTRUMENTS & TECHNIQUES

1. Laboratory safety and good laboratory practices 2. Principles and application of Laboratory instruments-microscope, incubator, autoclave, centrifuge, LAF, filtration unit, shaker, pH meter. 3. Buffer preparation & titration 4. Cleaning and Sterilization of glassware 5. Preparation of media- Nutrient Agar and Broth 6. Inoculation and culturing of bacteria in Nutrient agar and nutrient broth 7. Preparation of agar slant, stab, agar plate 8. Phenol Coefficient method to test the efficacy of disinfectants Herbarium: Plant collecting, Preservation and Documentation: Stepwise Practicing Herbarium techniques.

FIELD EQUIPMENTS

1. Global Positioning System (GPS) instrument & Collection of any wild 25 plant specimens
2. Learn to handle Herbarium making tools
3. Pressing and Drying of collected plant specimens
4. Special treatments for all varied groups of plants
5. Mount on standard herbarium sheets
6. Labelling with Standard method
7. Organize them and give Index Register Number



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8. Introduction and Importance of Bioprospecting

9. Bioreactors Instrumentation and herbal Preparations 1. Develop Capsules of herbs/, Develop Herbal oils/, Develop Poultice/cream Analyze some active ingredients using chromatography /Spectrophotometry 2. Microbial Techniques & instrumentation Microscopy – Light, phase contrast, electron, scanning and transmission electron microscopy, staining techniques for light microscopy, sample preparation for electron microscopy. Common equipment of microbiology lab and principle of their working – autoclave, oven, laminar air flow, centrifuge. Colorimetry and spectrophotometry, immobilization methods, fermentation, and fermenters. 3. Biopiracy, Biopatent, Intellectual Property Rights,

Reference Books:

1. Dawson, C. (2002). Practical research methods. UBS Publishers, New Delhi.
2. Stapleton, P., Yondeowei, A., Mukanyange, J., Houten, H. (1995). Scientific writing for agricultural research scientists – a training reference manual. West Africa Rice Development Association, Hong Kong.



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Subject Code	Subject title	Credit
MJL13BOT	COMPUTATIONAL BIOLOGY & RESEARCH METHODOLOGY FOR PLANTS-Lab	3

PRACTICALS:

1. Experiments based on chemical calculations.
2. Plant microtechnique experiments.
3. The art of imaging of samples through microphotography and field photography.
4. Poster presentation on defined topics.
5. Technical writing on topics assigned.



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Subject Code	Subject title	Credit
MJ14BOT	WATER RELATIONS, GROWTH AND DEVELOPMENT	3

Unit-1

Water relations of plants: Unique physio-chemical properties of water, chemical potential , water potential. Apparent free space, bulk movement of water, soil plant atmosphere, contuum (SPAC), stomatal regulation of transpiration, hormonal and energy dependent hypothesis; Inorganic nutrition, physicochemical aspects of solute transport, diffusion and facilitated diffusion, passive and active transport; Nernst equation and Donnan's potential. Role of ATPase as a carrier, co-transport (symport) and counter transport (antiport). Ion channels, role of calmodulin. Importance of foliar nutrition and use of chelates.

Unit-2

Photosynthesis: Energy pathway in photosynthesis, chloroplast as an energy transducing organelle; Composition and characterization of photo systems, I and II , electron flow through cyclic, noncyclic and pseudo cyclic photophosphorylation. Pathways of CO₂ fixation. Differences between C₃ and C₄ fixation and different kinds of C₄ pathways.

Unit-3

CAM pathway: Occurrence, biological events and adaptive advantage; Photorespiration: Mechanism and regulation of photorespiration; Introductory studies on water stress and its tolerance mechanisms.

Unit-4

Enzymes: Classification, mode of action, Km value; Industrial application, immobilized enzymes, their preparation and application; Enzyme regulation: Competitive and noncompetitive, allosteric enzymes.

Unit-5

Chemical control of growth and morphogenesis; Hormonal effects on growth and development; Bioassay of plant growth regulators and mode of action with reference to auxins; Gibberellins, cytokinins, abscisic acid and ethylene; Phytochrome: Chemistry and photo morphogenetic



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effects and role in flowering; Genetic study of secondary metabolites such as alkaloids (only 0 types of wide occurrence.) Dormancy: Seed and bud dormancy; hormonal regulation.

References:

1. V K Gupta, T S Paul. Fungi and Plant disease.
2. Malhotra, Aggarwal Ashok. Plant Pathology.
3. Rangaswamy, A Mahadevan. Diseases of crop plants in India.
4. B P Pandey. Plant Pathology.
5. George N Agrios (2006). Plant pathology (V Edn). Elsevier Academic Press



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Subject Code	Subject title	Credit
MJL14BOT	WATER RELATIONS, GROWTH AND DEVELOPMENT-LAB	1

PRACTICALS

1. Study of physical and chemical characteristics of soil by rapid field test.
2. Determination of pH of water.
3. Determination of dissolved oxygen in water
4. Extraction of amylase and determination of its activity
5. Preparation of cleared whole mounts of floral parts of polypetalae, sympetalae and monocots for vasculature.
6. With the help hand section and dissection prepare longitudinal and transverse sections of Flower. Examination of: a. Transmitting tissue/ canal in stigma and style. Use of paraffin Method of micro technique .
7. Acquaintance with ultratome: use of wood microtome and common and anatomy and histochemical methods.



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Subject Code	Subject title	Credit
MJ15BOT	ECOLOGY - CLIMATOLOGY, SOIL SCIENCE AND AUTECOLOGY	4

Unit-1

Definition, scope and concept of plant ecology; History of ecology and relation of ecology with other disciplines. Principles of ecology; Concept of environment, habitat and ecological niche. The environment we live in.

Unit-2

Light and temperature as ecological factors; Precipitation and Relative Humidity as ecological factors; Measurement and analysis of light, temperature, Precipitation and relation Humidity; Importance of water as an important factor on the life of plants.

Unit-3

Origin, development and formation of soil. Soil profile; Classification of soil; Effects of soil environment of plants; Chief soil types of India.

Unit-4

Biotic components of an ecosystem; Interrelation of various organisms; Population ecology, Natality, Mortality, Age distribution; Concept of carrying capacity.

Unit-5

Morphological, anatomical and physiological relation of plants with their environment; Plant indicators; Ecotypic and Ecadic differentiation; Physical and physiological dryness; Genecology

References:

1. V K Gupta, T S Paul. Fungi and Plant disease.
2. Malhotra, Aggarwal Ashok. Plant Pathology.
3. Rangaswamy, A Mahadevan. Diseases of crop plants in India.
4. B P Pandey. Plant Pathology.
5. George N Agrios (2006). Plant pathology (V Edn). Elsevier Academic Press.



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Subject Code	Subject title	Credit
MJL15BOT	ECOLOGY - CLIMATOLOGY, SOIL SCIENCE AND AUTECOLOGY-LAB	1

PRACTICALS

1. Learning techniques of making temporary and permanent microscopic preparation.
2. Knowledge and use of photomicrography in anatomical studies.
3. Knowledge and use of the principles and working of electron microscope
4. Learning to use simple experimental method in anatomical studies.
5. Determination of moisture constant of soil.
6. Determination of pH of water.
7. Determination of dissolved oxygen in water
8. Determination of following data. a. Solar energy b. Atmospheric temperature c. Relative Humidity
9. Determination of soil profile.



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SEMESTER VII

Subject Code	Subject title	Credit
MJ16BOT	BIOCHEMISTRY AND METABOLISM IN PLANTS	3

Unit-1

Lipid: General structure, classification; Synthesis of fatty acid; β -Oxidation; Synthesis of carbohydrates from fatty acids; Protein: Amino acid structure and Biosynthesis; protein conformation, protein synthesis.

Unit-2

Secondary plant products: General structure of important phenolic compound groups in plants; Shikimic acid pathway and phenolic compound synthesis; General structure and synthesis of alkaloids from amino acids; Growth regulators: structure and biosynthesis of Ethylene, Jasmonates and Brassinolides.

Unit-3

Cell wall components; Cellulose: structure and model for biosynthesis, structure and operation of the model for synthesis at plasma membrane level; Lignin: structure and biosynthesis in plants. Cyanogenesis: a general account of cyanogenic compounds as glycosides.

Unit-4

DNA and RNA: structure; DNA replication; Transcription in prokaryotes and eukaryotes, transcript modification; Regulation of translation.

Unit-5

Nitrate metabolism in plants; Nitrogen fixation, Nitrogenase system; Ammonium assimilation. GS-GOGAT system;



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References:

1. Jaibala, S. and Balakrishnan, G. (1975). A Hand book of common remedies based On Siddha
1. System of Indian medicines. St. Louis Institute Press, Chennai.
2. Natkarni, K.M. (1998). Indian Materia Medica (Vol. I-III). Popular Prakasam, New Delhi.
3. Raychaudri, S.P. (1991). Recent advances in Medicinal, Aromatic and Spice crops (Vol. I).
4. Today & Tomorrow publication, New Delhi.
5. Johnson,T (1999). CRC Ethnobotany desk Reference , CRC press ,New York



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Subject Code	Subject title	Credit
MJL16BOT	BIOCHEMISTRY AND METABOLISM IN PLANTS-LAB	1

Practicals:

1. Estimation of total nitrogen by kjaldahl method.
2. Principles of colorimetry, spectrophotometry and fluorimetry.
3. Estimation of titrable and total acidity.
4. Estimation of protein by Biuret and Lowry's method.



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B.Sc. Botany

Subject Code	Subject title	Credit
MJ17BOT	PLANT BREEDING	3

Unit-1

Introduction: Objectives of plant breeding, important achievements and future prospects. Genetic variability and its role in plant breeding. Domestication and centers of origin of cultivated plants.

Unit-2

Systems of reproduction in plants: Reproductive systems and pollination control mechanisms; Sexual reproduction - Cross and self-pollination; asexual reproduction, Incompatibility and Male Sterility, their types.

Unit-3

Hybridization: Hybridization - role and methods, Inter-varietal, inter specific and inter generic crosses. Back-cross breeding. Heterosis, Inbreeding depression.

Unit-4

Breeding for resistance: Breeding for biotic (disease) and abiotic (drought) stresses; loss due to diseases, disease development, disease escape, disease resistance, vertical and horizontal resistances of biotic stress; methods of breeding for disease resistance.

Unit-5

Mutation breeding: Mutagens and crop improvement. Spontaneous and induced mutations, effects of mutation. Physical and chemical mutagens; principles and working of Gamma gardens, methods of mutation breeding, mutations in oligogenic traits, mutations in polygenic traits, limitations of mutation breeding, achievements of mutation breeding. Role of mutations in Plant Breeding; Modern trends in plant breeding.



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References

1. Allard R W (1995). Principles of Plant Breeding. John Wiley and Sons, Inc.
2. Ghahal G S and Gosal S S (2002). Principles and procedures of Plant Breeding. Narosa Publishing House.
3. Sharma J R (1994). Principles and practices of Plant Breeding. Tata McGraw-Hill Publishers Company Ltd.
4. Singh B D (1996). Plant Breeding: Principles and methods. Kalyani Publications.



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B.Sc. Botany

Subject Code	Subject title	Credit
MJL17BOT	PLANT BREEDING-LAB	1

Practicals:

1. Isolation and culture of protoplast.
2. Isolation and screening of industrially important microorganism.
3. Isolation of plant DNA, plasmid DNA, bacteriophage DNA.



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B.Sc. Botany

Subject Code	Subject title	Credit
MJ18BOT	PLANT BIOTECHNOLOGY: IN VITRO CULTURE, GENETIC ENGINEERING AND IPR ISSUE	3

Unit-1

Concept and scope of Biotechnology; Techniques of tissue culture, cell culture and organ culture; Sterilization, culture media; In-vitro auxorrophs, disease resistance, salt and drought resistance, nutritional quality and herbicide resistance.

Unit-2

Micropropagation; Production of haploids: anther culture and pollen culture Somatic embryogenesis, somaclonal variation; Protoplast culture: isolation, culture and fusion of protoplast; IPR-general idea about patents. Copyright, trademark and geographical indication.

Unit-3

Biotransformation: production of useful compounds through cell culture; factors affecting yield: bioreactors; Strategies of microbial strain improvement; The recombinant DNA concept and principle of cloning; Isolation and purification of DNA.

Unit-4

Restriction endonuclease : properties and types; Blotting southern, northern and western; Selection and screening of recombinant clone; Cloning vehicles salient features: plasmid , cosmid & Tiplasmid.

Unit-5

Single stranded DNA viruses CaMV Lambda phage vectors M13 vectors; Expression vectors; Cloning construction of genomic and DNA libraries; Application of- r- DNA technology in plant improvement.



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References:

1. Jaibala, S. and Balakrishnan, G. (1975). A Hand book of common remedies based On Siddha
1. system of Indian medicines. St. Louis Institute Press, Chennai.
2. Natkarni, K.M. (1998). Indian Materia Medica (Vol. I-III). Popular Prakasam, New
Delhi.
3. Raychaudri, S.P. (1991). Recent advances in Medicinal, Aromatic and Spice crops (Vol.
I).
4. Today & Tomorrow publication, New Delhi.
5. Johnson,T (1999). CRC Ethnobotany desk Reference , CRC press ,New York



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B.Sc. Botany

Subject Code	Subject title	Credit
MJL18BOT	PLANT BIOTECHNOLOGY: IN VITRO CULTURE, GENETIC ENGINEERING AND IPR ISSUE-LAB	1

1. Separation of DNA by agarose gel electrophoresis -
2. Estimation of RNA by Orcinol method.
3. Separation of amino acids by paper chromatography.
4. Separation of serum proteins by paper electrophoresis.
5. Thin layer chromatography of mycotoxins
6. SDS-Page of proteins.
7. Performance of affinity chromatography.
8. Performance of Gel filtration chromatography.
9. Demonstration of blotting technique.[any one].
10. Ion exchange chromatography



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B.Sc. Botany

Subject Code	Subject title	Credit
MJ19BOT	HERBAL TECHNOLOGY, INDUSTRIAL MICROBIOLOGY	3

Unit-1

History of Medicinal plants. Traditional Medicinal systems: Ayurvedha, Siddha, Unani and Naturopathy. Definition of Drug - Classification of natural drugs, (Alphabetical, Morphological, Pharmacological, Chemical and Chemo taxonomical). Traditional and Folklore medicines - Native medicine.

Unit -2

Pharmacognosy - Definition and Scope. Drug adulteration, Drug evaluation, Chemical evaluation, Physical evaluation and Biological evaluation. Phytochemical investigations,

Unit -3

Cultivation, collection and preparation of natural drugs - Macroscopic characters (Physical and Organoleptic characters), therapeutical and pharmaceutical uses of the following medicinal plants: Adathoda vasica, Aloe vera, Centella asiatica, Syzygium aramoticum, Datura metel, Piper nigrum, Allium sativum, Azadiracta indica, Ocimum sanctum and Vinca rosea.

Unit –4

Industrial Microbiology- Industrial product of vinegar & citric acid. Industrial product of antibiotics; penicillin and streptomycin Microbes as a source of Single Cell protein (SCP). Mushrooms Dairy product from microorganisms; butter, yogurt and cheese. Biopesticides, Biofertilizer, Biopolymers, Bioremediation.

Unit –5

Ethnobotany – definition-Major tribes of South India and their ethno botanical and ethnobiological heritage. Ethno Medicines. Ethnobotany and conservation of plants with special reference to India – mythology and conservation of ecosystems, conservation of selected plant species: sacred groves, forestry and unique ecosystems and their ethnobiological values, plants and animals in art, tradition and ethnography: Ethnobotanical field methods.



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B.Sc. Botany

Suggested Books:

1. John Jothi Prakash, E. (2003). Medicinal Botany and Pharmacognosy. Page 41 of 96
JPR Publication, Vallioor, Tirunelveli.
2. Gokhale, S.B., Kokate, C.K. and Purohit, A.P. (2003). Pharmacognosy. Nirali
Prakashan, Pune.
3. Prajapathi, Purohit, Sharma and Kumar. (2003). A Hand book of Medicinal plants.
Agrobios, Publications, Jodhpur.



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B.Sc. Botany

Subject Code	Subject title	Credit
MJL19BOT	HERBAL TECHNOLOGY, INDUSTRIAL MICROBIOLOGY-LAB	1

1. Study of typical compound microscope
2. Micrometry measurement of a fungal spore
3. Determination of cell density by counting chamber
4. Isolation of bacteria from soil, air and water
5. Isolation of fungi from soil
6. Contact slide technique
7. Streak and Pour plate techniques
8. Simple staining of a bacterium
9. Negative staining of a bacterium
10. Gram staining — Positive and negative

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B.Sc. Botany

SEMESTER VIII

Subject Code	Subject title	Credit
MJ20BOT	BIO INFORMATICS	3

UNIT I

Introduction to computers and bioinformatics - Types of operating systems, concepts of networking and remotelogs, basic fundamentals of working with unix.

UNIT II

Biological databases- Overview, modes of database search , mode of data storage (Flat file format, db-tables), flat file formats of Gen Bank, EMBL, DDBJ, PDB.

UNIT III

Sequence alignment —Concept of local and global sequence alignment, Pairwise sequence alignment, scoring alignment, substitution matrices, multiple sequence alignment. UNIT IV. Phylogenetic analysis- Basic concepts of phylogenetic analysis, rooted/unrooted trees, approaches for phylogenetic tree construction (UPGMA, Neighbor joining, Maximum parsimony, and Maximum likelihood).

UNIT VI

Generation and analysis of high throughput sequence data – Assembly pipeline for clustering of HTGS data, format of “.ace” file, quality assessment of genomic assemblies, International norms for sequence data quality, Clustering of EST sequences, concept of Unigene. Annotation procedures for high through-put sequence data- Identification of various genomic elements (protein coding genes, repeat elements, strategies for annotation of whole genome, functional annotation of EST clusters, gene ontology (GO) consortium.

UNIT VII

Structure predictions for nucleic acids and proteins - Approaches for the prediction of RNA secondary and tertiary predictions, energy minimization and base covariance models, Basic approaches for protein structure predictions, comparative modeling, fold recognition/“threading” and ab-initio prediction.



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B.Sc. Botany

Essential Reading

- a. Gary B. Fogel & David Corne, Evolutionary Computation in Bioinformatics, Morgan Kaufmann Publishers,
- b. Christoph W. Sensen, Essentials Of Genomics and Bioinformatics, Wiley-VCH.
- c. David Bowtell and Joseph Sambrook, DNA Microarrays, CSHL Press
- d. Murray R. Selwyn, Principles of Experimental Design for the Life Sciences, CRC Press
- e. Warren J. Ewens, Gregory R. Grant, Statistical Methods in Bioinformatics : An Introduction, Springer.
- f. Hugh G. Griffin, Annette M. Griffin, PCR Technology: current innovation.
- g. Ben Hui Liu, Statistical Genomics : Linkage, mapping and QTL Analysis, CRC Press
- h. Westhead, Instant Notes on Bioinformatics, 1st Edn.
- i. Baxevanis & Francis, Bioinformatics, 2nd Edn. Wiley.



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Subject Code	Subject title	Credit
MJL20BOT	BIO INFORMATICS-LAB	1

1. Use of NCBI Bioinformatics Tools:

- a) Pubmed b) OMIM c) Taxonomy
- d) Protein analysis e) Genes and Disease

2. Use of Expasy Tools

3. FASTA and sequence formats

4. BLAST

5. Alignments – Pair wise & global

6. Construction of Dendrogram

7. Prediction of ORF/ gene Prediction

8. Protein Visualization (RASMOL, SPDB VIEWER, PROTEINEXPLORER)

9. Protein Modeling-Homology modeling & Active site Prediction

10. Primer Designing

11. Biostatistics: Mean, Median, Mode, Standard Deviation, Chi square test, Student's t-test



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Subject Code	Subject title	Credit
AMJ01BOT	ADVANCE MOLECULAR BIOLOGY	3

Unit 1: Introduction to Molecular Cloning

Vectors: Characteristics of cloning vectors, Plasmids (pBR322, pUC18/I9) and Ti plasmid. Shuttle vectors and Expression vectors: E.coli lac and T7 promoter-based vectors. Enzymes used in Molecular Cloning: Restriction enzymes. Types I, II and III, nomenclature, use of Type II restriction enzymes in cloning. Reverse transcriptase.

Methods used in Molecular Cloning: Agarose gel electrophoresis of DNA, Southern, Northern and Western blotting. RFLP (Restriction Fragment Length Polymorphism). Molecular probes: cDNA probes – RNA probes

Unit 2: PCR Techniques

Principle of Polymerase Chain Reaction, RT-PCR, Real-Time PCR and their applications. (10 lectures)

Unit 3: Gene Expression

Regulation of gene expression in Prokaryotes: various models - operon - details of lac operon- negative and positive control lac operon

Regulation gene expression in eukaryotes: Regulation of transcription - regulation of RNA processing and translation. Microarray and gene expression analysis.

Unit 4: DNA Sequencing

DNA sequencing: Maxam Gilbert chemical method - Sanger's enzymatic chain termination method – foot printing.



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Unit 5: Gene Silencing and Genome Editing

Introduction to gene silencing (RNAi)/ post-transcriptional gene silencing (PTGS) and its mechanism. Introduction and Principle of genome editing

Reference Books:

1. Brown TA. (2010) Gene Cloning and DNA Analysis. 6th edition. Blackwell Publishing, Oxford, U.K.
2. Primrose SB and Twyman RM. (2006) Principles of Gene Manipulation and Genomics, 7th edition. Blackwell Publishing, Oxford, U.K.
2. Sambrook J and Russell D. (2001) Molecular Cloning-A Laboratory Manual. 3rd edition. Cold Spring Harbor Laboratory Press.
3. Walker J M and Gringold EB, Molecular Biology and Biotechnology. Panima.
4. Benjamin Lewin. Genes 1X. John Wiley.
5. Hartwell L H et al., Genetics: From Genes to Genome. Mc Graw Hill.
6. Watson J D et al., Molecular Biology of the Gene. The Benjamin / Cummings.
7. Lodish H et al., Molecular Cell Biology. Scientific American Books. W H Freeman.
8. David Freid felder, Molecular Biology. Narosa.
9. Adrin J Harwood, Methods in Molecular Biology, Vol.58, Basic DNA and RNA protocols. Humana Press.



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B.Sc. Botany

Subject Code	Subject title	Credit
AMJL01	ADVANCE MOLECULAR BIOLOGY-Lab	1

PRACTICALS:

1. Isolation of plasmid DNA.
2. Agarose Gel Electrophoresis of plasmid DNA.
3. Preparation of competent cells for transformation by calcium chloride method.
4. Transformation of E.coli host cell with plasmid DNA.
5. Digestion of plasmid DNA using restriction enzymes and analysis by agarose gel electrophoresis.



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Subject Code	Subject title	Credit
AMJ02BOT	NANOBIOTECHNOLOGY	3

NANO-BIOTECHNOLOGY

1. Introduction of Nano biotechnology and its applications. Various types of nanomaterial utilized in agriculture.
2. Nanoparticles in agricultural and food diagnostics: Enzyme Biosensors and Diagnostics – DNA Based Biosensors and Diagnostics, Radiofrequency Identification.
3. Nanotechnology in food production: Food and new ways of food production -Efficient fractionation of crops, Efficient product structuring -Optimizing Nutritional Values - Applications of Nanotechnology in Foods: Sensing, Engineering Food Ingredients to Improve Bioavailability - Nanocrystalline Food Ingredients – Nano-emulsions – Nano Engineered Protein Fibrils as Ingredient Building Blocks.
4. Nanotechnology in food packaging: Reasons to Package Food Products. Smart nanomaterials for packaging.

Reference Books:

1. Xiong, Essential Bioinformatics. Cambridge University Press.
2. Marketa J Zvelebil, Understanding Bioinformatics. Garland Science.
3. Shui Quing Ye, Bioinformatics: A practical Approach.
4. Anna Tramontano, Introduction to Bioinformatics
5. David W Mount, Bioinformatics. CBS
6. Mani K and Vijayaraj N, Bioinformatics. Kalaikathir Achchagam.
7. Augen Jeff, Bioinformatics in the post genomic era. Addison Wesley.
8. The 2018-2023 World Outlook for Nanobiotechnology Paperback – December 18, 2017, Icon group international.
9. Arunava Goswami and Samrat Roy Choudhury, Nanobiotechnology, Basic and Applied Aspects.
10. Clive Jarvis, Nanobiotechnology: An Introduction.
11. H B Singh, S Mishra, L F Fraceto, R D D Lima; Emerging Trends in Agri-Nanotechnology



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B.Sc. Botany

Subject Code	Subject title	Credit
AMJL02BOT	NANOBIOTECHNOLOGY	1

PRACTICALS:

1. Search and Sequence retrieve from genbank database.
2. Alignment of sequence by using tools: Clustal X, Clustal W, Mega and Bioedit.
3. Phylogenetic tree analysis by using Mega software.
4. Primer designing by using online tools.



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B.Sc. Botany

Subject Code	Subject title	Credit
AMJ03BOT	APPLIED BOTANY	3

1. Role of microbes in Industries and Human Welfare

- Production and application of organic acids; lactic acid, citric acid and acetic acid. Concept of antibiosis, secondary metabolites, antibiotic fermentation.
- Biological wastewater treatment: Upflow Anaerobic Sludge Blanket (USAB), Reactor and Fluidized Bed Reactor (FBR).
- Food toxicology: Microbial toxins (Endotoxin and exotoxin). Source of microbial toxin in contamination of food grains and food products, spoilage of food.
- Basic concept in brief FDA (Food and Drug Administration), EPA (Environment Protection Act), HACCP (Hazard Analysis and Critical Control Points) and FSA (Flexible Spending Account).

2. Application of microbes in fermentation processes:

Types, design and maintenance of bioreactors. Application of fermentation technology in industry

3. Conventional fuels and their environmental impact

Microbial enhanced oil recovery, Bio-ethanol and bio-diesel production, commercial production from lignocellulosic waste, Biogas production – Methane and hydrogen production using microbial culture. Extremophiles and their biotechnological applications.

4. Production of antibiotics, vaccines, and biocides.

Bioreactors; Bioprocess engineering; Production of non-microbial origin products by genetically engineered microorganisms. Concept of probiotics and applications of new tools of biotechnology for quality feed/food production. Single cell protein, Bioinsecticides; Biofertilizers; Recent advances in microbial biotechnology. Mass cultivation of Spirulina, Chlorella and Scenedesmus, Commercial potential of Spirulina, Dunaliella and Porphyra.



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5. A brief account on Phytochemical and Pharmacological aspects:

A brief account on Phytochemical and Pharmacological aspects and uses of following medicinal plants: *Andrographis paniculate*, *Bacopa monnieri*, *Centella asiatica*, *Curcuma longa*, *Momordica charantia*, *Ocimum sanctum*, *Phyllanthus niruri*, *Tinospora cordifolia*, *Withania somnifera*.

6. Conventional versus non-conventional methods for crop improvement.

Genetic engineering for resistance against abiotic and biotic stresses; Genetic engineering for increasing crop productivity; Genetic engineering for quality improvement. Molecular breeding: constructing molecular maps, physical and molecular maps; diversity assessment and phylogenetic analysis; molecular tagging of genes/traits.

7. Classical ways of genome analysis.

DNA chips and their use in transcriptome analysis. General uses and application of Crystallography. Genomics and proteomics of cyanobacteria, yeast and *Fusarium*. Applications of genomics and proteomics in agriculture, human health and industry.

Reference Books:

1. Kamp RM. 2004. *Methods in Proteome and Protein Analysis*. Springer.
2. Primrose SB & Twyman RM. 2007. *Principles of Genome Analysis and Genomics*. Blackwell.
3. Sensen CW. 2005. *Handbook of Genome Research*. Vols. I, II. Wiley CVH.
4. Jordening H-J & Winter J. 2006. *Environmental Biotechnology: Concepts and Applications*. Wiley-VCH Verlag.
5. Aggarwal Rattan Lal. *Fundamentals of Plant Breeding and Hybrid Seed Production*.



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B.Sc. Botany

Subject Code	Subject title	Credit
AMJL03BOT	APPLIED BOTANY -Lab	1

PRACTICALS:

1. Enrichment culture and isolation of agriculturally important microorganisms
2. Isolation of antibiotic producing microorganisms
3. Isolation of industrially important microorganisms, their maintenance and improvement.
4. Production of industrial compounds such as industrial alcohol/ citric acid/lactic acid and their recovery.
5. Study of bio-reactors and their operations.
6. Demonstration of bioinsecticides / biofertilizers production.
7. Media preparation, culture/cultivation of Spirulina
8. Isolation of natural products by column chromatography
9. Separation and Isolation of natural products by Preparative TLC.
10. Micro/Macro-Algae isolation technique form local algal flora.



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Subject Code	Subject title	Credit
RC01	RESEARCH METHODOLOGY	4

1. Foundations of Research: Meaning, Objectives, Motivation, Utility. Concept of theory, empiricism, deductive and inductive theory. Characteristics of scientific method - Understanding the language of Research - Concept, Construct, Definition, Variable. Research Process.

2. Problem Identification & Formulation - Research Question - Investigation Question - Measurement Issues - Hypothesis - Qualities of a good Hypothesis. Null Hypothesis & Alternative Hypothesis. Hypothesis Testing - Logic & Importance.

3. Research Design: Concept and Importance in Research - Features of a good research design - Exploratory Research Design - concept, types and uses, Descriptive Research Designs - concept, types and uses. Experimental Design: Concept of Independent & Dependent variables.

4. Qualitative and Quantitative Research: Qualitative research - Quantitative research - Concept of measurement, causality, generalization, replication. Merging the two approaches.

5. Measurement: Concept of measurement- what is measured? Problems in measurement in research- Validity and Reliability. Levels of measurement Nominal, Ordinal, Interval, Ratio.

6. Sampling: Concepts of Statistical Population, Sample, Sampling Frame, Sampling Error, Sample Size, Non Response. Characteristics of a good sample. Probability Sample- Simple Random Sample, Systematic Sample, Stratified Random Sample & Multi-stage sampling. Determining size of the sample Practical considerations in sampling and sample size.



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7. Data Analysis: Data Preparation - Univariate analysis (frequency tables, bar charts, pie charts, percentages), Bivariate analysis- Cross tabulations and Chisquare test including testing hypothesis of association.

8. Interpretation of Data and Paper Writing- Layout of a Research Paper, Journals in Computer Science, Impact factor of Journals, When and where to publish ? Ethical issues related to publishing, Plagiarism and Self-Plagiarism.

9. Use of Encyclopedias, Research Guides, Handbook etc., Academic Databases for Computer Science Discipline.

10. Use of tools & techniques for Research: methods to search required information effectively, Reference Management Software like Zotero/ Mendeley, Software for paper formatting like LaTeX/ MS Office, Software for detection of Plagiarism

Reference Books:

1. Business Research Methods- Donald Cooper & Pamela Schindler, TMGH, 9th editions.
2. Business Research Methods- Alan Bryman & Emma Bell, Oxford University Press.
3. Research Methodology- C. R. Kothari
4. Garg, B.L., Karadia, R., Agarwal, F. and Agarwal, U.K., 2002. An introduction to Research Methodology, RBSA Publishers.
5. Sinha, S.C. and Dhiman, A.K., 2002. Research Methodology, Ess Ess Publications. 2 volumes.
6. Trochim, W.M.K., 2005. Research Methods: the concise knowledge base, Atomic Dog Publishing. 270p.



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Subject Code	Subject title	Credit
RC02	RESEARCH PROPOSAL	4

Process of writing a research proposal.

That includes the following points:

1. TITLE.
2. INTRODUCTION
3. REVIEW OF LITERATURE (BACKGROUND AND RATIONALE)
4. RESEARCH QUESTION(S) ..
5. AIMS & OBJECTIVES
6. RESEARCH METHODOLOGY.
7. PLAN OF WORK
8. REFERENCE/BIBLIOGRAPHY.



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Subject Code	Subject title	Credit
RC03	Research Report	4

A research report is a reliable source to recount details about conducted research. It is most often considered to be a true testimony of all the work done to garner specificities of research.

Research reports present the results of formal investigations into the properties, behavior, structures, and principles of material and conceptual entities. Almost any physical phenomenon or concept may be investigated in a research framework. The following are some key differences between formal research, and other less structured kinds of inquiry.

1. *Problem definition*: the rigorous reduction of the inquiry to a narrow question with a quantifiable answer. The most significant preliminary phase of research writing is that of effective problem definition. This process is one of identifying an interesting question and narrowing the research inquiry to a manageable size.
2. *Research approach*: the structuring of the research according to a methodology associated with a specialized field of inquiry. Specialized fields have research methodologies that are followed in investigating problems. These range from general methods of interviewing and literature researching to highly specialized procedures for using materials and mechanical devices to establish appropriate conditions for generating data. Adapting a sound research methodology to the investigation of your problem is a major milestone in the conduct of your inquiry.
3. *Research report*: the presentation of the research and its results in a rigorously formatted document that follows a conventional structure. In presenting your research, you pull all its elements together into a focused, coherent document. Research reports contain a standard set of elements that include.



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B.Sc. Botany

MINOR (I/ III/ V/ VII)

SEMESTER I

Subject Code	Subject title	Credit
MN01BOT	Botany I (Minor)	4

- 1. Introductory Botany:** - Elementary idea of classification of plant kingdom, Taxonomic hierarchy up to species, Botanical Nomenclature of plants, herbarium - types and use. Family description- Solanaceae and Poaceae. A concise introduction to branches and scope of botany.
- 2. Diversity of Plants:** General identifying features, structures, life cycle and economic importance of the following groups: a) Bacteria b) Virus c) Algae d) Fungi e) Lichens f) Bryophyta g) Pteridophyta h) Gymnosperms i) Angiosperms. Morphology of flowering plants (Monocots and Dicots). What is Ethnobotany and its scope? Brief notes on ethnobotanical uses of the following plants with particular reference to Jharkhand a. Kalmegh b. Palash c. Karanj d. Neem e. Bamboo f. Sal
- 3. Plant pathology-** What is plant pathology its scope and benefits , Symptoms and control of the following diseases: a. Wilt of tomato b. Citrus canker c. Yellow vein mosaic of bhindi d. Leaf curl of papaya e. Black stem rust of wheat f. Loose smut of wheat
- 4. Plant physiology:** - Ascent of Sap, Transpiration, Photosynthesis and Respiration. Introductory knowledge of pollination, Fertilization and seed development in Angiosperms, Monocot and Dicot seeds; Germination of seeds. Biological Nitrogen fixation.

Reference Books

1. Botany for degree students; A.C. Dutta
2. College Botany; Vol I, Ganguly, Das and Dutta
3. College Botany; Vol. II, Ganguly, Kar and Santra
4. Study of Botany; Mitra, Mitra and Guha
5. A text book of Botany; K. S. Bilgrami
6. A text book of Botany; Vol. I & II, Hait, Bhattacharya and Ghosh
7. Practical botany: Bendre and Kumar, and S. P. Lal



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B.Sc. Botany

Practical

1. Family descriptions of given plant
2. Anatomical features of: monocot (root and stem) and dicot (root and stem).
3. Gram's staining technique.
4. Botanical name, common name, family and uses of cereal, pulses, oil yielding.
5. Acquaintance with the instruments.
6. Identifying features and name of the given materials/permanent slides (algae, fungi, bryophytes, pteridophytes and gymnosperms).
7. Mitosis and Meiosis: Study of materials/permanent slides.



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B.Sc. Botany

SEMESTER III

Subject Code	Subject title	Credit
MN03BOT	Botany II (Minor)	4

Plant taxonomy: - Introduction to plant taxonomy Identification, Classification, Nomenclature. Identification Functions of Herbarium, important herbaria and botanical gardens of the world and India. Elementary idea of documentation of Flora. Botanical nomenclature Principles and rules (ICBN); ranks and names; binominal system, typification, author citation, valid publication, rejection of names, principle of priority and its limitations.

Plant anatomy: -Meristematic and permanent tissues Root and shoot apical meristems; Simple and complex tissues. Organs Structure of dicot and monocot root stem and leaf. Secondary Growth Vascular cambium – structure and function, seasonal activity. Secondary growth in root and stem, Wood (heartwood and sapwood). Adaptive and protective systems Epidermis, cuticle, stomata; General account of adaptations in xerophytes and hydrophytes.

Plant embryology: - Structural organization of flower Structure of anther and pollen; Structure and types of ovules; Types of embryo sacs, organization and ultrastructure of mature embryo sac. Pollination and fertilization Pollination mechanisms and adaptations; Double fertilization; Seed-structure. Embryo and endosperm Endosperm types, structure and functions; Dicot and monocot embryo. Apomixis and polyembryony Definition, types and practical applications.

Plant Ecology: - Introduction General Introduction. Ecological factors Climatic, Edaphic and Biotic factors: Variation of Optimal and limiting factors; Shelford law of tolerance. Ecological Adaptation of hydrophytes and xerophytes. Plant communities Succession (Primary and secondary); hydrosere, xerosere. Major vegetation in world.



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Reference Books

1. Simpson, M.G. (2006). Plant Systematics. Elsevier Academic Press, San Diego, CA, U.S.A.
2. Singh, G. (2012). Plant Systematics: Theory and Practice. Oxford & IBH Pvt. Ltd., New Delhi. 3rd edition.
3. Bhojwani, S.S. & Bhatnagar, S.P. (2011). Embryology of Angiosperms. Vikas Publication House Pvt. Ltd. New Delhi. 5th edition.
4. Mauseth, J.D. (1988). Plant Anatomy. The Benjamin/Cummings Publisher, USA.

Practical

1. Making herbarium and Mounting of a properly dried and pressed specimen of any wild plant with herbarium label (to be submitted in the record book).
2. Study of meristems through permanent slides and photographs.
3. Study of Tissues (parenchyma, collenchyma and sclerenchyma); Macerated xylary elements, Phloem (Permanent slides, photographs)
4. Stem: Monocot: Zea mays; Dicot: Helianthus; Secondary: Helianthus (only Permanent slides).
5. Study of Structure of anther (young and mature), tapetum (amoeboid and secretory) (Permanent slides).
6. Study of Types of ovules: anatropous, orthotropous, circinotropous, amphitropous/campylotropous.
7. Dissection of embryo/endosperm from developing seeds.



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B.Sc. Botany

SEMESTER V

Subject Code	Subject title	Credit
MN05BOT	ANGIOSPERM ANATOMY, EMBRYOLOGY AND PALYNOLOGY	3

Unit-1

Origin, growth, differentiation and ultra structure of cell and tissue, fine structure of plasmodesmata, microtubules, microfibrils and secondary structure; Apical, lateral and intercalary meristems- their ultra structure and histochemistry, organogenesis; Ontogeny, phylogeny, ultra structure and function of primary and secondary xylem; wood anatomy;

Ontogeny, phylogeny, ultra structure and function of primary and secondary phloem; Structure variability in leaves, leaf histogenesis, leaf meristem, origin, development ultra structure of trichomes and stomata.

Unit-2

Nodal anatomy-nodal types and evolutionary consideration, Vascular cambium vs. cork cambium factors controlling their activity, periderm, lenticles, abscission, wound healing; Anatomy of monocotyledons and dicotyledonous seed and fruits, seed appendages, their anatomy structure and function; Anatomy in relation to taxonomy; Contemporary plant anatomy: current trends and prospects.

Unit-3

Microsporangium- structure and function of wall layers, ultra structure change in tapetum and meiocytes during Microsporogenesis, role of tapetum, pollen development, anther culture and haploid plants; Pollen wall morphogenesis-microspore pollen mitosis; division of generative cell; pollen fertility and sterility; pollen storage viability and germination; Megasporeogenesis, various types of embryosacs, their development and fertilization.



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Unit-4

Embryology and taxonomy; diagnostic embryological characters, primitive and advanced characters, comparative embryology of hybrids dysfunction of endosperm, arrested development of embryo.

Unit-5

Development and evolution of pollen types; stereo and ultrastructure of exine, apertures, furrow. Palynology and taxonomy; Aerobiology and its application. Aeropalynology, methods of aerospora survey and analysis, pollen allergy and pollen calendars system approach for allergy. Mellitopalyonolgy: general account Paleopalynolgy: role in coal and oilgenesis.



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SEMESTER V

Subject Code	Subject title	Credit
MNL05BOT	ANGIOSPERM ANATOMY, EMBRYOLOGY AND PALYNOLOGY-LAB	1

1. Acquaintance with ultratomy: use of wood microtomy and common and anatomy and Histochemical methods.
2. Learning techniques of making temporary and permanent microscopic preparation.
3. Knowledge and use of photomicrography in anatomical studies.
4. Knowledge and use of the principles and working of electron microscope
5. Learning to use simple experimental method in anatomical studies.



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SEMESTER VII

Subject Code	Subject title	Credit
MN07BOT	PLANT BIOTECHNOLOGY	3

Unit 1: Plant Tissue Culture Introduction, Composition of media; Nutrient and hormone requirements (role of vitamins and hormones); Totipotency; Organogenesis; Embryogenesis (somatic and zygotic); Protoplast isolation, culture and fusion; Tissue culture applications (micropropagation, androgenesis, virus elimination, secondary metabolite production, haploids, triploids and hybrids; Cryopreservation; Germplasm Conservation).

Unit 2: Recombinant DNA technology Restriction Endonucleases (History, Types I-IV, biological role and application); Restriction Mapping (Linear and Circular); Cloning Vectors: Prokaryotic (pUC 18 and pUC19, pBR322, Ti plasmid, BAC); Lambda phage, M13 phagemid, Cosmid, Shuttle vector; Eukaryotic Vectors (YAC).

Unit 3: Gene Cloning Recombinant DNA, Bacterial Transformation and selection of recombinant clones, PCR mediated gene cloning; Gene Construct; construction of genomic and cDNA libraries, screening DNA libraries to obtain gene of interest by genetic selection; complementation, colony hybridization; PCR.

Unit 4: Methods of gene transfer Biological method (Indirect): Agrobacterium-mediated; Physical methods (Direct): Electroporation, Microinjection, Microprojectile bombardment; Selection of transgenics– selectable marker and reporter genes (Luciferase, GUS, GFP).

Unit 5: Applications of Biotechnology Pest resistant (Bt-cotton); herbicide resistant plants (RoundUp Ready soybean); Transgenic crops with improved quality traits (Flavr Savr tomato, Golden rice); Improved horticultural varieties (Moondust carnations); Role of transgenics in bioremediation (Superbug); edible vaccines; Industrial enzymes (Aspergillase, Protease, Lipase); Genetically Engineered Products–Human Growth Hormone; Humulin; Biosafety concerns.



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Reference Books:

1. Bhojwani, S.S. and Razdan, M.K., (1996). Plant Tissue Culture: Theory and Practice. Elsevier Science Amsterdam. The Netherlands.
2. Glick, B.R., Pasternak, J.J. (2003). Molecular Biotechnology- Principles and Applications of recombinant DNA. ASM Press, Washington.
3. Bhojwani, S.S. and Bhatnagar, S.P. (2011). The Embryology of Angiosperms. Vikas Publication House Pvt. Ltd., New Delhi. 5th edition.
4. Snustad, D.P. and Simmons, M.J. (2010). Principles of Genetics. John Wiley and Sons, U.K. 5th edition.
5. Stewart, C.N. Jr. (2008). Plant Biotechnology & Genetics: Principles, Techniques and Applications. John Wiley & Sons Inc. U.S.A.



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Subject Code	Subject title	Credit
MNL07BOT	PLANT BIOTECHNOLOGY	1

PRACTICALS:

1. (a) Preparation of MS medium. (b) Demonstration of in vitro sterilization and inoculation methods using leaf and nodalex plants of tobacco, Datura, Brassica etc.
2. Study of anther, embryo and endosperm culture, micropropagation, somatic embryogenesis & artificial seeds through photographs.
3. Isolation of protoplasts.
4. Construction of restriction map of circular and linear DNA from the data provided.
5. Study of methods of gene transfer through photographs: Agrobacterium-mediated, direct gene transfer by electroporation, microinjection, microprojectile bombardment.
6. Study of steps of genetic engineering for production of Bt cotton, Golden rice, FlavrSavr tomato through photographs.
7. Isolation of plasmid DNA.
8. Restriction digestion and gel electrophoresis of plasmid DNA.
